PHYS 598: Physics of Living Systems

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1. The external environment of cells

Where we discuss physical challenges cells face, before discussing what a cell is

Before we discuss what a cell even is, let's begin our course by using only pure thought and math to deduce the types of challenges a cell would face from its external environment. As we will see, the cell must infer a changing environment from finite molecular encounters, divide finite surface resources among many tasks, and distinguish desired signals from decoy signals generated by competing molecules in the outside environment. From probability, diffusion, optimization, and thermodynamics, we will derive design constraints on a cell without yet specifying what a cell is.

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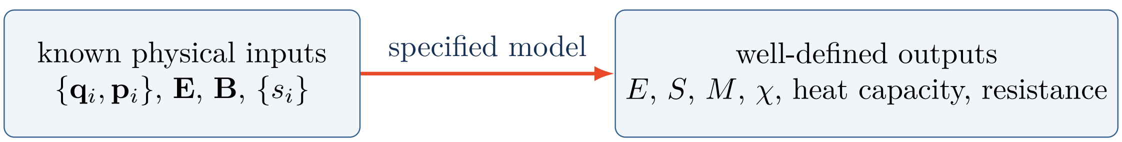
1 Why biophysics is hard

1.1 In physics for non-living matter, numbers map to numbers

For a nonliving system, we often begin with quantities whose physical meaning is already clear: positions $\{\mathbf{q}_i\}$, momenta $\{\mathbf{p}_i\}$, electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, or spins $\{s_i\}$. A specified model maps these inputs to other well-defined quantities: energy, entropy, magnetization, susceptibility, heat capacity, conductivity, resistance, or an order parameter.

Even when calculating the mapping is very hard, the question is usually recognizable: we know what the variables mean and what the desired output is. Statistical mechanics and condensed-matter physics may classify superconductors, insulators, magnets, and other phases of non-living systems with quantities that have operational definitions and reproducible procedures for computing or measuring them.



The mapping can be difficult to solve while still being clear to state.

Figure 1: Typical procedure for non-living physics: defined inputs (numbers or mathematical functions), a specified mapping from those inputs to outputs, and defined outputs (also numbers or mathematical functions).

1.2 Living systems: requires a map from numbers to behaviors

For a living system, we may start with numbers or quantifiable features too: a DNA sequence, the abundance of a particular protein, a concentration of a biomolecule, or a mechanical force on the cell. But with these numerical inputs, the desired outputs are often higher-order functions or behaviors that defy mathematical functions. These include:

- Divide or do not divide
- Die or live
- Become a neuron or another cell type
- Consume one type of sugar instead of another
- Coordinate a collective decision with neighboring cells

The difficulty is not merely that the mapping from input (quantities) to output (behavior) is complicated. Often, we do not even know, a priori, which quantifiable features of a cell are useful as inputs of a model, how they could be coarse-grained, or what even the form of dynamical systems is. The quantifiable features can be specific to each biomolecule of a cell and largely non-mechanical, thereby not fitting easily into the language of established physics (e.g., momenta, energy, etc.). Hence, we might know the behavior we want to explain, but we might need to reverse-engineer both the relevant inputs and the map.

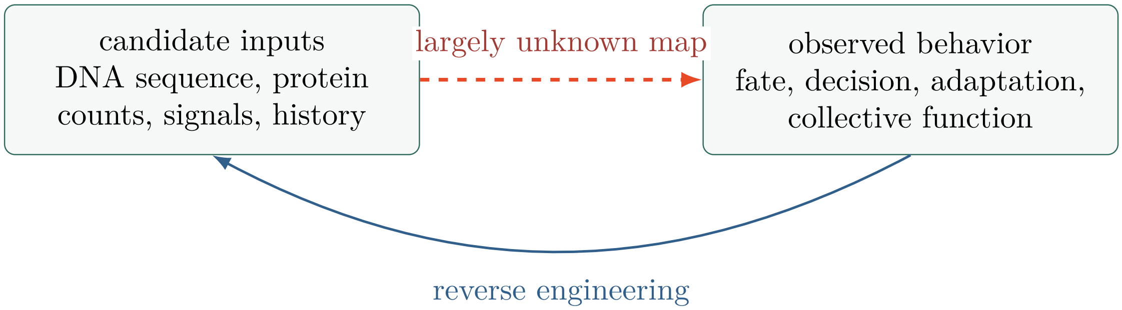


Figure 2: Challenge in biophysics: the behavior (output) is observable but hard to quantify as a mathematical function. Moreover, the relevant input variables and the mapping from those inputs to the behavior are often unknown.

Key question

Think of a cellular behavior that sounds qualitative. What numerical observable would convince two independent researchers that the behavior occurred? Would that observable explain the behavior, or only label it?

2 Without knowing what a cell is, can we deduce what it must accomplish?

Before discussing what a cell is, consider its external environment — this is called the 'extracellular environment.' A cell would need to sense the molecules present in its extracellular environment using some type of sensors. Not having "eyes" like us, the primary mode of sensing for a cell is simple molecular encounter: a diffusing molecule in the extracellular environment finds a sensor and binds to it.

Suppose the environment contains a chemical at concentration $\bar{c}$. The environment is larger than the cell and the detectors on the cell. Even if $\bar{c}$ is perfectly constant on average, molecules move randomly. A small detector therefore sees a fluctuating number of molecules. The cell must infer a stable environmental quantity from those fluctuating encounters, and it must do so within a finite decision time T.

For simplicity, we won't consider a cell here but just consider a physical detector. As another simplicity, we'll assume that it's just a cube with edges of length L. After the derivations below, we could later let this detector represent a whole cell, a molecular sensor on the surface of a cell, or a detector inside a cell (e.g., some region on DNA where a molecule binds).

The key question we will answer is:

$$\textit{How small can the fractional uncertainty } \frac{\mathrm{sd}(\hat{c})}{\bar{c}} \textit{ be?}$$

Here, sd is the standard deviation in $\hat{c}$. To answer it, we first need to quickly review binomial distribution.

3 Mathematical interlude: the binomial distribution

3.1 Bernoulli trials and counting

Consider N independent trials. Trial i produces a Bernoulli random variable

$$X_i = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } q = 1 - p. \end{cases} \tag{1}$$

The total number of successes is

$$Z_N = \sum_{i=1}^{N} X_i. \tag{2}$$

Then Z_N is binomially distributed:

$$P(Z_N = n) = \binom{N}{n} p^n q^{N-n}, \qquad n = 0, 1, \ldots, N. \tag{3}$$

The coefficient $\binom{N}{n}$ counts which n of the N trials succeeded. $p^n q^{N-n}$ is the probability for any one such pattern.

3.2 Three ways to obtain the mean

Method 1: Intuitively, if a coin with success probability p is tossed N times, the expected fraction of successes is p. The expected number is therefore Np.

Method 2: From linearity of expectation,

$$\langle Z_N \rangle = \sum_{i=1}^{N} \langle X_i \rangle = \sum_{i=1}^{N} (1 \cdot p + 0 \cdot q) = Np. \tag{4}$$

Method 3: Starting from equation (3),

$$\begin{aligned} \langle Z_N \rangle &= \sum_{n=0}^{N} n \binom{N}{n} p^n q^{N-n} && (5) \\ &= Np \sum_{n=1}^{N} \binom{N-1}{n-1} p^{n-1} q^{(N-1)-(n-1)} && (6) \\ &= Np. && (7) \end{aligned}$$

The second line uses $n\binom{N}{n} = N\binom{N-1}{n-1}$. The remaining sum is the normalization of a binomial distribution with $N - 1$ trials.

3.3 Variance

For one Bernoulli trial, $\langle X_i \rangle = p$, so

$$\begin{aligned} \mathrm{Var}(X_i) &= \left\langle (X_i - p)^2 \right\rangle && (8) \\ &= p(1-p)^2 + q(0-p)^2 && (9) \\ &= pq^2 + p^2 q = pq. && (10) \end{aligned}$$

For independent trials, the variances add. Therefore

$$\boxed{\langle Z_N \rangle = Np, \qquad \mathrm{Var}(Z_N) = Npq.} \tag{11}$$

Direct check: Using $n^2 = n(n-1) + n$ and $n(n-1)\binom{N}{n} = N(N-1)\binom{N-2}{n-2}$,

$$\langle Z_N(Z_{N-1}) \rangle = \sum_{n=0}^{N} n(n-1)\binom{N}{n} p^n q^{N-n} = N(N-1)p^2, \tag{12}$$

$$\left\langle Z_N^2 \right\rangle = N(N-1)p^2 + Np, \tag{13}$$

$$\mathrm{Var}(Z_N) = \left\langle Z_N^2 \right\rangle - \langle Z_N \rangle^2 = Np(1-p) = Npq. \tag{14}$$

4 Refresher on statistics

The next results are general, not limited to the binomial distribution. They require finite variances and carefully stated assumptions about correlations.

4.1 Variance of a sum

Let $\mu_X \equiv \langle X \rangle$ and $\mu_Y \equiv \langle Y \rangle$. Then, we have

$$\begin{aligned} \mathrm{Var}(X+Y) &= \left\langle \left((X-\mu_X) + (Y-\mu_Y)\right)^2 \right\rangle & (15)\\ &= \left\langle (X-\mu_X)^2 \right\rangle + \left\langle (Y-\mu_Y)^2 \right\rangle + 2\left\langle (X-\mu_X)(Y-\mu_Y) \right\rangle & (16)\\ &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y), & (17) \end{aligned}$$

where

$$\mathrm{Cov}(X,Y) = \langle (X - \langle X \rangle)(Y - \langle Y \rangle) \rangle\,. \tag{18}$$

If X and Y are independent, their covariance vanishes, giving

$$\boxed{\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \quad \text{for independent or uncorrelated } X, Y.} \tag{19}$$

For N identically distributed independent variables,

$$\begin{aligned} \mathrm{Var}\left(\sum_{i=1}^{N} X_i\right) &= \sum_{i=1}^{N} \mathrm{Var}(X_i) & (20)\\ &= N\,\mathrm{Var}(X_1). & (21) \end{aligned}$$

This gives $\mathrm{Var}(Z_N) = Npq$ when the X_i are Bernoulli distributed.

4.2 Why the standard error of the mean (SEM) falls as $1/\sqrt{M}$

Let $X_1, \ldots, X_M$ be independent measurements of the same random variable, each with mean μ and variance σ^2. Their sample mean is

$$\bar{X}_M = \frac{1}{M}\sum_{i=1}^{M} X_i. \tag{22}$$

Using equation (21),

$$\mathrm{Var}(\bar{X}_M) = \frac{1}{M^2}\,\mathrm{Var}\left(\sum_{i=1}^{M} X_i\right) \tag{23}$$

$$= \frac{1}{M^2} M\sigma^2 = \frac{\sigma^2}{M}. \tag{24}$$

Taking the square root gives the standard error of the mean (SEM):

$$\boxed{\mathrm{SEM}(\bar{X}_M) = \mathrm{sd}(\bar{X}_M) = \frac{\sigma}{\sqrt{M}}.} \tag{25}$$

When σ is unknown and estimated by the sample standard deviation s, the reported estimate is $s/\sqrt{M}$.

5 The Berg–Purcell limit

In 1977, Howard Berg and Edward Purcell asked whether diffusion and molecular counting impose a physical limit on chemoreception [1]. We will derive the core scaling for a transparent cubic detector.

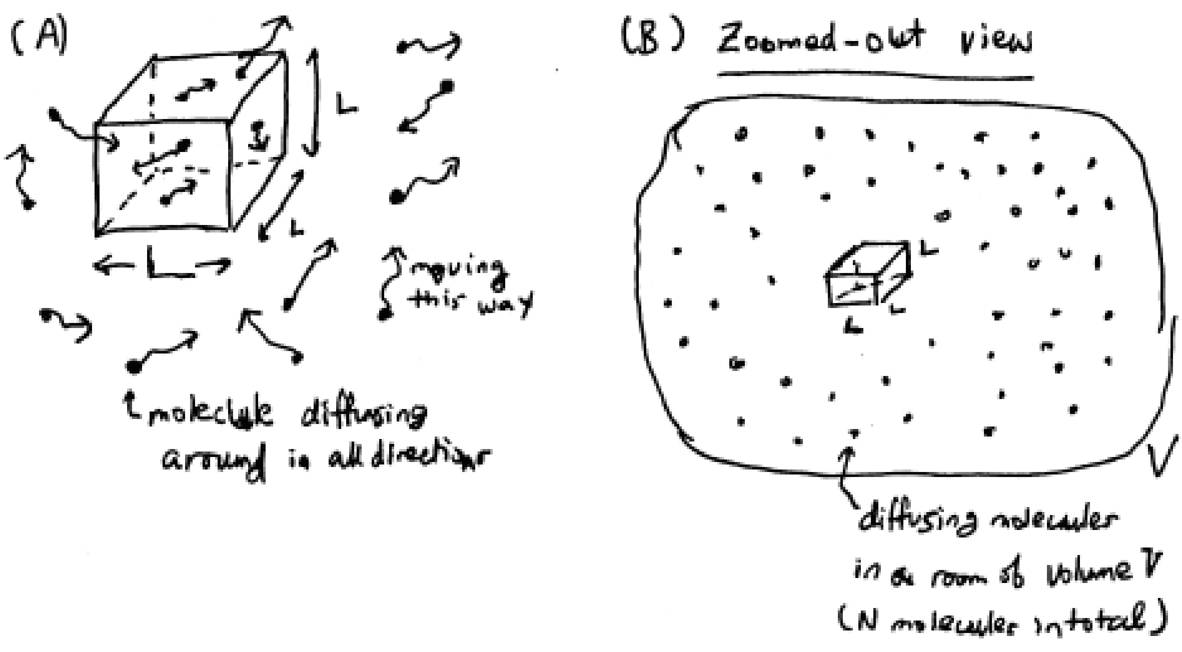


Figure 3: The Berg–Purcell setup. (A) Molecules diffuse through a permeable cubic detector of side L. (B) The detector occupies a small part of a bath of volume V containing N molecules. The detector reads the instantaneous number of molecules inside it.

Here are the relevant parameters:

Symbol	Units	Meaning
V	volume	volume of the large environment
N	number	total number of independently diffusing molecules
$\bar{c} = N/V$	number/volume	true mean concentration
$u = L^3$	volume	volume of the transparent cubic detector
D	length^2/time	molecular diffusion coefficient
T	time	time available to form a concentration estimate
n_i	number	molecules inside the detector in snapshot i
$\widehat{c}_i = n_i/u$	number/volume	concentration inferred from snapshot i

Table 1: Parameters for Berg–Purcell setup

We have only a few starting assumptions:

1. $V \gg u$, and the mean concentration is spatially uniform and remains constant during T.
2. Molecules do not interact and diffuse independently with diffusion coefficient D;
3. The detector does not perturb the molecules;
4. A snapshot reports the instantaneous number of molecules inside the detector;
5. The final estimate is an average of the snapshots collected over T.

6 Towards Berg–Purcell limit: one snapshot

For any one molecule, the probability of being inside the detector in a snapshot is the fraction of the bath occupied by the detector:

$$p = \frac{u}{V}, \qquad q = 1 - \frac{u}{V}. \tag{26}$$

Because each of the N molecules supplies an independent inside/outside trial,

$$n_i \sim \mathrm{Binomial}\left(N, \frac{u}{V}\right). \tag{27}$$

The inferred concentration in snapshot i is $\widehat{c}_i = n_i/u$. Its mean is

$$\langle \widehat{c}_i \rangle = \frac{\langle n_i \rangle}{u} = \frac{N(u/V)}{u} = \frac{N}{V} = \bar{c}. \tag{28}$$

Thus the estimator is unbiased under the model. Its variance is

$$\mathrm{Var}(\widehat{c}_i) = \frac{1}{u^2}\,\mathrm{Var}(n_i) \tag{29}$$

$$= \frac{1}{u^2} N \frac{u}{V}\left(1 - \frac{u}{V}\right) \tag{30}$$

$$= \frac{\bar{c}}{u}\left(1 - \frac{u}{V}\right). \tag{31}$$

For a detector much smaller than the bath,

$$\mathrm{Var}(\widehat{c}_i) \simeq \frac{\bar{c}}{u}. \tag{32}$$

Hence, the fractional standard deviation from one-snapshot measurement by the detector is

$$\frac{\mathrm{sd}(\widehat{c}_i)}{\bar{c}} \simeq \frac{1}{\sqrt{\bar{c}u}} = \frac{1}{\sqrt{\bar{c}L^3}}. \tag{33}$$

7 Towards Berg–Purcell limit: Repeated measurements in time

7.1 Why snapshots taken sufficiently close in time are correlated

Suppose the detector takes two snapshots separated by a picosecond. Nearly all molecules that were inside during the first snapshot remain inside during the second. The detector has mostly recounted the same molecules. It has not obtained a second independent measurement.

Diffusion refreshes the occupancy inside the box. That is, if we wait long enough, the original molecules inside the box now will have diffused out of the box. We just need to calculate how long we must wait for this to occur. We can use dimensional argument here. Specifically, in three dimensions,

$$\left\langle |\Delta \mathbf{r}(t)|^2 \right\rangle = 6Dt. \tag{34}$$

For an order-of-magnitude derivation, the detector loses memory of a configuration when molecules diffuse a distance comparable to L. Hence

$$\tau_{\text{corr}} \sim \frac{L^2}{D}. \tag{35}$$

This is a decorrelation timescale, not a guaranteed moment at which every old molecule has left. Diffusive trajectories have a broad range of exit and return times.

The effective number of independent measurements available in time T is then

$$M_{\text{eff}} \sim \frac{T}{\tau_{\text{corr}}} \sim \frac{DT}{L^2}. \tag{36}$$

7.2 Average the independent information

Let the time-averaged estimate be

$$\widehat{c}_T = \frac{1}{M_{\text{eff}}} \sum_{i=1}^{M_{\text{eff}}} \widehat{c}_i. \tag{37}$$

Combining the one-snapshot error in equation (33) with the SEM result in equation (25),

$$\frac{\operatorname{sd}(\widehat{c}_T)}{\bar{c}} \sim \frac{1}{\sqrt{\bar{c}L^3}} \frac{1}{\sqrt{DT/L^2}} \tag{38}$$

$$= \frac{1}{\sqrt{\bar{c}DTL}}. \tag{39}$$

Because the argument estimated rather than exactly calculated the temporal correlations, we write the Berg–Purcell form as

$$\boxed{\frac{\operatorname{sd}(\widehat{c}_T)}{\bar{c}} \sim \frac{\alpha}{\sqrt{\bar{c}DTL}}} \qquad \text{(Berg–Purcell sensing form).} \tag{40}$$

Here α is a dimensionless order-one constant. For a fully specified detector and estimator, the relation can be made precise. More generally it is a characteristic error scale. Calling it a rigorous lower bound requires a specified sensing model, estimator, and set of available resources.

7.3 What the scaling says

The denominator in equation (40) is dimensionless:

$$[\bar{c}DTL] = L^{-3}(L^2T^{-1})TL = 1. \tag{41}$$

The result predicts:

- more molecules ($\bar{c}\uparrow$) reduce counting noise.
- faster diffusion ($D\uparrow$) delivers fresh information more rapidly.
- longer integration ($T\uparrow$) permits more independent information.
- a larger detector ($L\uparrow$) counts more molecules, although its occupancy also refreshes more slowly.

A scaling estimate

Suppose $\bar{c} = 1\,\text{molecule}/\mu\text{m}^3$, $D = 100\,\mu\text{m}^2/\text{s}$, $L = 1\,\mu\text{m}$, and $T = 1\,\text{s}$. Then $\bar{c}DTL = 100$, so the fractional error scale is $\alpha/10$.

7.4 Same scaling, different sensing schemes

Different detectors treat molecules differently. A transparent monitor can recount molecules that leave and return. A perfectly absorbing detector removes each molecule after counting it. A reversible receptor on a surface of a cell can switches between bound and unbound states and can be unavailable while occupied. Receptor arrays and downstream readouts introduce further correlations and resource constraints.

For ideal spherical detectors of radius a, exact calculations give different coefficients while retaining the same $1/(cDaT)$ variance scaling [2, 3]:

Sensing scheme	Fractional variance	α if $L = a$
Transparent monitoring sphere	$\dfrac{3}{5\pi DacT}$	$\sqrt{\dfrac{3}{5\pi}} \simeq 0.437$
Perfectly absorbing sphere	$\dfrac{1}{4\pi DacT}$	$\dfrac{1}{\sqrt{4\pi}} \simeq 0.282$

Table 2: Two ideal detectors with the same scaling and different prefactors. Absorption prevents rebinding and recounting.

For a reversible single receptor, diffusion, receptor occupancy, intrinsic association, and rebinding all contribute. Particle-based theory and simulations resolve these correlations and add a reaction-noise term [4, 5]. Collective sensing provides another example: the communication architecture between cells changes the coefficient because their measurements become correlated [6]. A broader synthesis by ten Wolde, Mugler, and colleagues shows how receptors, integration time, readout molecules, and energy constrain sensing [7].

8 A changing environment: when old measurements become misleading

The Berg–Purcell calculation assumed that the concentration was stationary during the measurement time T. If the environment changes, measurements from the distant past may be statistically independent but no longer relevant to the present. We can use the same type of reasoning we used for deriving the Berg-Purcell limit to derive a similar limit – a lower bound on fractional error in concentration determination – for a changing environment. Below, we consider a simple scenario of a changing environment.

8.1 A local ramp and a causal average

Let $t = 0$ denote the present. Over a sufficiently short interval, any smooth concentration trajectory can be approximated by

$$c(t) = c_0 + vt, \qquad -T \leq t \leq 0, \tag{42}$$

where $c_0 = c(0)$ is the concentration now and $v = \dot{c}(0)$ is its local rate of change. A simple causal estimator averages the measurements available during the preceding T seconds:

$$\widehat{c}_T = \frac{1}{T}\int_{-T}^{0} \widehat{c}(t)\ \mathrm{d}t. \tag{43}$$

It is causal because it uses the past but not the future.

Suppose the instantaneous measurement noise has zero mean. Then

$$\langle \widehat{c}_T \rangle = \frac{1}{T}\int_{-T}^{0} c(t)\ \mathrm{d}t \tag{44}$$

$$= \frac{1}{T}\int_{-T}^{0} (c_0 + vt)\ \mathrm{d}t \tag{45}$$

$$= c_0 - \frac{vT}{2}. \tag{46}$$

Thus $\widehat{c}_T$ is an unbiased estimate of the *past average*, but a biased estimate of the concentration now. Its bias is

$$b(T) \equiv \langle \widehat{c}_T \rangle - c_0 = -\frac{vT}{2}. \tag{47}$$

Define the environmental-change timescale

$$\tau_{\mathrm{env}} \equiv \frac{c_0}{|v|}. \tag{48}$$

It is the time over which the concentration would change by an amount comparable to itself if the local rate persisted. The squared fractional bias is

$$\frac{b(T)^2}{c_0^2} = \frac{T^2}{4\tau_{\mathrm{env}}^2}. \tag{49}$$

8.2 Noise plus bias gives an optimal memory

Over a window short enough that $c(t)$ remains near c_0, use the stationary Berg–Purcell variance as a local approximation:

$$\frac{\mathrm{Var}(\widehat{c}_T)}{c_0^2} \simeq \frac{\alpha^2}{c_0 DLT}. \tag{50}$$

It is convenient to define a molecular sampling timescale

$$\tau_{\mathrm{s}} \equiv \frac{\alpha^2}{c_0 DL}. \tag{51}$$

Because $c_0 DL$ has units of inverse time, τ_{s} has units of time. The fractional sampling variance is simply τ_{s}/T.

Therefore the fractional mean-squared error for estimating the present value is

$$\boxed{\mathcal{E}(T) \equiv \frac{\left\langle (\widehat{c}_T - c_0)^2 \right\rangle}{c_0^2} \simeq \frac{\tau_{\mathrm{s}}}{T} + \frac{T^2}{4\tau_{\mathrm{env}}^2}.} \tag{52}$$

The first term decreases with T: longer averaging suppresses counting noise. The second increases with T: longer averaging gives more weight to stale information.

Differentiate with respect to T:

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}T} = -\frac{\tau_{\mathrm{s}}}{T^2} + \frac{T}{2\tau_{\mathrm{env}}^2}. \tag{53}$$

Setting this derivative to zero gives

$$\boxed{T_* = \left(2\tau_{\mathrm{s}}\tau_{\mathrm{env}}^2\right)^{1/3} = \left(\frac{2\alpha^2\tau_{\mathrm{env}}^2}{c_0 DL}\right)^{1/3}.} \tag{54}$$

The second derivative is positive, so this stationary point is a minimum. At the optimum,

$$\frac{\tau_{\mathrm{s}}}{T_*} = 2\frac{T_*^2}{4\tau_{\mathrm{env}}^2}. \tag{55}$$

The sampling variance is twice the squared staleness bias. The two errors are comparable but not equal because they have different powers of T. The minimum fractional mean-squared error is

$$\mathcal{E}_{\mathrm{min}} = \frac{3}{2^{4/3}}\left(\frac{\tau_{\mathrm{s}}}{\tau_{\mathrm{env}}}\right)^{2/3}. \tag{56}$$

Accurate tracking requires a separation of timescales. Ideally,

$$\frac{L^2}{D} \ll T_* \ll \tau_{\mathrm{env}}. \tag{57}$$

The first inequality permits many decorrelation times within the measurement. The second makes the local ramp and the nearly stationary noise approximation self-consistent.

9 Finite geometry: a finite-sized detector cannot measure everything equally well

The preceding calculations gave the detector one sensing task. A bounded object, having a finite size, faces a second challenge: it has finite volume and finite boundary area. Even before specifying anything about its operation, its geometry physically limits the number of objects it can contain and the number of detectors it can place on its boundary.

9.1 Finite volume turns concentrations into finite copy numbers

For a compartment of volume V_c, a molar concentration c_{mol} corresponds to a mean copy number

$$\bar{N}_c = c_{mol} V_c N_A, \tag{58}$$

where N_A is Avogadro's constant. In convenient units,

$$\boxed{\bar{N}_c \simeq 0.602 \left(\frac{c_{mol}}{\text{nM}}\right)\left(\frac{V_c}{\text{fL}}\right).} \tag{59}$$

Thus 1 nM in 1 fL corresponds to substantially less than one molecule on average. This does not mean that a fraction of a molecule is present. It means that repeated snapshots often contain zero molecules and occasionally contain one or more. In an ideal dilute-reservoir model with a Poisson count,

$$P(N_c = 0) = e^{-\bar{N}_c}. \tag{60}$$

For $\bar{N}_c = 0.602$, the probability of zero is about 0.55.

Key idea: concentration is an average, but copy number is discrete

For large copy numbers, a smooth concentration field can be a sufficient description. But at order-one copy number, the same mean concentration describes a probability distribution over integer-valued counts. Here, a small volume of a detector (typical for cells) could give rise to a misleading belief that every molecular species must be scarce. This tells us that we must take discreteness of molecules, their integer counts, seriously for biological processes, rather than relying just on concentrations.

9.2 A finite boundary supplies a finite number of detector slots

Use a sphere of radius R as a reference geometry:

$$A = 4\pi R^2, \qquad V_c = \frac{4\pi R^3}{3}, \qquad \frac{A}{V_c} = \frac{3}{R}. \tag{61}$$

Suppose only a fraction ϕ of the boundary can be devoted to the sensing tasks under discussion. If detector type i has footprint area a_i and copy number R_i, then

$$\boxed{\sum_{i=1}^{K} a_i R_i \le A_s, \qquad A_s \equiv \phi A.} \tag{62}$$

The rest of the boundary can be reserved for transport, mechanics, or other functions that we have not yet named. For equal footprints $a_i = a$, the total number of available detector slots is

$$N_R = \frac{A_s}{a}. \tag{63}$$

At fixed detector density, $N_R \propto R^2$, whereas volume grows as R^3. Thus the number of boundary detectors per unit volume decreases as $1/R$.

10 Specificity: the target must compete against every decoy

A detector may bind the molecule of interest, but it may also bind other molecules. The relevant question is not merely whether the target molecule binds to its receptor. Also important is the question of whether the detector can distinguish target binding from all competing molecules that are not the target (i.e., different molecular species that can still bind the receptor).

Equilibrium binding: mass action, with an optional statistical-mechanics interpretation

The mass-action route below is sufficient for the key results. The later statistical-mechanics route is left as an optional self-study part.

Mass-action route: receptor bookkeeping. Let T denote the target molecule (i.e., molecule = ligand) and let D_j, with $j = 1, \ldots, J$, denote the jth decoy ligand. In this section the subscript T means "target," not the measurement time used earlier. First label all possible ligand species generically as $L_1, L_2, \ldots$. Consider the simplest detector: one receptor R with one binding site. The receptor's mutually exclusive states are

$$R, \qquad RL_1, \qquad RL_2, \qquad \ldots$$

It cannot be unbound and bound at the same time, nor can this one-site receptor bind two ligands simultaneously. Conservation of receptors therefore gives

$$[R]_{\text{tot}} = [R] + \sum_i [RL_i]. \tag{64}$$

For each ligand species,

$$R + L_i \rightleftharpoons RL_i, \tag{65}$$

and the dissociation constant is defined by

$$K_{d,i} = \frac{[R][L_i]}{[RL_i]}. \tag{66}$$

Writing the *free* ligand concentration as $c_i = [L_i]$ and rearranging gives

$$[RL_i] = [R]\frac{c_i}{K_{d,i}}, \qquad \frac{[RL_i]}{[R]} = \frac{c_i}{K_{d,i}} \equiv x_i. \tag{67}$$

Thus x_i is the population of receptors bound to species i, measured relative to the unbound-receptor population.

Substituting this result into receptor conservation yields

$$\begin{aligned} [R]_{\text{tot}} &= [R] + \sum_i [R]\frac{c_i}{K_{d,i}} \\ &= [R]\left(1 + \sum_i \frac{c_i}{K_{d,i}}\right). \end{aligned} \tag{68}$$

Dividing by $[R]$ defines

$$\boxed{Z \equiv \frac{[R]_{\text{tot}}}{[R]} = 1 + \sum_i \frac{c_i}{K_{d,i}} = 1 + \sum_i x_i.} \tag{69}$$

This is the partition function for the one-receptor model, also called its *binding polynomial*.

From this expression, we see that:

- The 1 is the relative weight of the unbound state: $[R]/[R] = 1$. It does not mean that there is only one receptor.
- The weights are added because the states are mutually exclusive alternatives. Independent sites that could be occupied simultaneously would instead produce additional combined states.
- Both c_i and $K_{d,i}$ have concentration units, so $x_i = c_i/K_{d,i}$ and Z are dimensionless.
- The concentration c_i already represents the abundance of species i. The index i runs over ligand species, not over individual ligand molecules. A ligand can compete through high abundance, small K_d, or both.

The concentrations in these equations are free concentrations. If receptor binding appreciably depletes a ligand pool, its free concentration must first be found using ligand mass conservation.

At equilibrium, the fraction of receptor copies in a state—equivalently, the probability that a receptor sampled at a random time occupies that state—is its relative weight divided by the sum of all weights:

$$P(\text{unbound}) = \frac{1}{Z}, \qquad P(i \text{ bound}) = \frac{x_i}{Z} = \frac{c_i/K_{d,i}}{Z}. \tag{70}$$

These probabilities sum to one by construction.

As a sanity check, suppose only one ligand species is present. Then

$$Z = 1 + \frac{c}{K_d}, \qquad P(\text{bound}) = \frac{c/K_d}{1 + c/K_d} = \frac{c}{c + K_d}. \tag{71}$$

This is the familiar Langmuir binding curve. In particular, $c = K_d$ gives $P(\text{bound}) = 1/2$: the dissociation constant is the free ligand concentration at which this simple one-site receptor is half occupied.

One target and many decoys: Now specialize the generic ligand labels to one target T and J decoys $D_1, \ldots, D_J$. Their binding reactions are

$$R + T \rightleftharpoons RT, \qquad R + D_j \rightleftharpoons RD_j. \tag{72}$$

Let c_T and $c_{D,j}$ be their free concentrations, and let $K_{d,T}$ and K_{d,D_j} be their dissociation constants. Mass action gives the relative weights

$$\begin{aligned} x_T &\equiv \frac{[RT]}{[R]} = \frac{c_T}{K_{d,T}}, \\ x_{D,j} &\equiv \frac{[RD_j]}{[R]} = \frac{c_{D,j}}{K_{d,D_j}}, \qquad W_D \equiv \sum_{j=1}^{J} x_{D,j}. \end{aligned} \tag{73}$$

Here W_D is the combined weight of every decoy species. Receptor conservation becomes

$$\begin{aligned} [R]_{\text{tot}} &= [R] + [RT] + \sum_{j=1}^{J} [RD_j] \\ &= [R](1 + x_T + W_D), \\ Z &= 1 + x_T + W_D. \end{aligned} \tag{74}$$

In the probabilities below, the event T means that a randomly sampled receptor is bound by the target, while "any decoy" means that it is bound by one of the D_j. Therefore

$$\begin{aligned} P(T) &= \frac{x_T}{Z}, \qquad & P(D_j) &= \frac{x_{D,j}}{Z}, \\ P(\text{any decoy}) &= \frac{W_D}{Z}, \quad & P(\text{occupied}) &= \frac{x_T + W_D}{Z}. \end{aligned} \tag{75}$$

Conditional on the receptor being occupied, the unbound state is excluded and the remaining probabilities must be renormalized:

$$P(T \mid \text{occupied}) = \frac{x_T}{x_T + W_D}, \qquad \epsilon \equiv P(\text{any decoy} \mid \text{occupied}) = \frac{W_D}{x_T + W_D}. \tag{76}$$

The wrong-to-right odds are consequently

$$\frac{P(\text{any decoy} \mid \text{occupied})}{P(T \mid \text{occupied})} = \frac{W_D}{x_T} = \frac{\sum_{j=1}^{J} c_{D,j}/K_{d,D_j}}{c_T/K_{d,T}}. \tag{77}$$

For J distinct decoy species with the same concentration c_D and dissociation constant $K_{d,D}$, this reduces to

$$\boxed{\frac{P(\text{any decoy} \mid \text{occupied})}{P(T \mid \text{occupied})} = J\frac{c_D}{c_T}\frac{K_{d,T}}{K_{d,D}}.} \tag{78}$$

A smaller K_d means stronger binding. Thus the target may bind more strongly than each individual decoy, $K_{d,T} < K_{d,D}$, yet sufficiently many decoy species—or sufficiently abundant decoys—can collectively dominate.

Moral of the mass-action calculation. Specificity is not solely a property of the receptor–target pair. It depends on the target's weight relative to the *sum* of the weights of every competitor in the surrounding molecular environment. Affinity and abundance enter together through c/K_d.

Notice also that $[R]_{\text{tot}}$ cancels from the conditional probabilities and from the wrong-to-right odds. Under this simple equilibrium model, adding more identical receptors produces more occupied receptors and more binding events and can therefore improve sensitivity. Sampling precision improves only to the extent that those events provide independent, or effectively independent, observations. In either case, adding receptors does not change the target-versus-decoy odds. This conclusion assumes that the receptors are identical and that binding does not appreciably deplete the free ligand concentrations. Improving equilibrium specificity requires changing the relative weights, or else using additional information such as binding duration and nonequilibrium proofreading.

Optional statistical-mechanics route. At equilibrium, a state with Gibbs free energy G_s has Boltzmann weight

$$e^{-\beta G_s}, \qquad \beta \equiv \frac{1}{k_B \Theta}, \tag{79}$$

where Θ is absolute temperature. We use Θ to distinguish temperature from both the measurement time T used earlier and the target label T used here. For two states,

$$\frac{P(a)}{P(b)} = e^{-\beta(G_a - G_b)}. \tag{80}$$

A lower free energy corresponds to a larger probability. An advantage of $5k_B\Theta$, for example, changes the odds by $e^5 \simeq 148$.

Let ΔG_i° be the standard Gibbs free-energy change for binding and c° the standard concentration. Thermodynamics gives

$$\Delta G_i^\circ = k_B \Theta \ln\left(\frac{K_{d,i}}{c^\circ}\right), \tag{81}$$

and therefore

$$x_i = \frac{c_i}{K_{d,i}} = \frac{c_i}{c^\circ} e^{-\beta \Delta G_i^\circ}. \tag{82}$$

The weight contains both molecular affinity and ligand abundance. A weaker binder can matter if enough copies are present.

Free-energy interpretation of the same competition. For the identical-decoy case, define the target's binding-free-energy advantage

$$\Delta\Delta G \equiv \Delta G_D^\circ - \Delta G_T^\circ > 0. \tag{83}$$

Because $K_{d,T}/K_{d,D} = e^{-\beta\Delta\Delta G}$,

$$\frac{P(\text{any decoy} \mid \text{occupied})}{P(T \mid \text{occupied})} = J\frac{c_D}{c_T}e^{-\beta\Delta\Delta G}. \tag{84}$$

Demanding $\epsilon \le \epsilon_0$ requires

$$\boxed{\Delta\Delta G \ge k_B\Theta\left[\ln\left(J\frac{c_D}{c_T}\right) + \ln\left(\frac{1-\epsilon_0}{\epsilon_0}\right)\right].} \tag{85}$$

Multiplying the number of decoy species by ten requires an additional free-energy advantage $k_B\Theta \ln 10$. This is an entropy-of-alternatives penalty.

For example, with $J = 1000$ equally abundant decoy species and target, requiring $\epsilon_0 = 0.05$ gives

$$\frac{\Delta\Delta G}{k_B\Theta} \ge \ln(1000) + \ln(19) \simeq 9.85. \tag{86}$$

A distribution of affinities. If the decoy binding free energies are independent Gaussian variables,

$$\Delta G_j^\circ \sim \mathcal{N}(\mu_D, \sigma_D^2), \tag{87}$$

then

$$\langle W_D \rangle = J\frac{c_D}{c^\circ}\exp\left(-\beta\mu_D + \frac{\beta^2\sigma_D^2}{2}\right). \tag{88}$$

Variability increases the expected decoy weight because the exponential gives disproportionate importance to the strongly binding tail. With many competitors, the strongest accidental binder can become an extreme-value problem [8].

11 Why all this is statistical physics

We still have not defined a cell, and we did not discuss any detailed terminologies from biology such as signaling pathway, a genome, an organelle, or any biochemical mechanisms. Instead, we discussed:

1. Discrete particles and counting statistics;
2. Diffusion as the mechanism that refreshes local information;
3. Causal estimation in a changing environment;
4. Geometry, finite resources, and constrained optimization;
5. Boltzmann weights and competition among microscopic states; and
6. Likelihood ratios for extracting identity from stochastic event durations.

These items or approaches are familiar to us from statistical mechanics in which we can derive entropy, fluctuations, or response constraints without tracking every microscopic detail of a system. Similarly, in this lecture, we have seen how we can derive sensing errors and design tradeoffs without knowing virtually anything about biology or a cell. Microscopic details – the specific molecules and their chemical properties – would define the α in the Berg-Purcell limit, the available area and allocation weights, and the ligand affinities and kinetic rates. But the form of the mathematical equations involved are set by physics.

12 Summary

In this lecture, we deduced, merely from pure thought and math – without knowing any biology – a set of problems that any proposed cell design must solve. Biology – the specific molecules making up a cell, their arrangements, and their chemistry – would lead us to deduce the mechanisms by which cells resolve these problems (or why a cell cannot resolve them). Physics provides the constraints obeyed by those solutions.

References and further reading

The original concentration-sensing argument is in Berg and Purcell [1]. Important studies that determined the exact prefactor for the Berg-Purcell limit under various sensing schemes include Endres and Wingreen [2], Mora and Wingreen [3], Bialek and Setayeshgar [4], Kaizu et al. [5], Fancher and Mugler [6], ten Wolde et al. [7]. Changing environments are treated in Mora and Nemenman [9], Malaguti and ten Wolde [10]. Resource allocation is developed in different settings by Govern and ten Wolde [11], Mayer et al. [12]. Specificity and crosstalk connect to Košmrlj et al. [8], Mora [13], Friedlander et al. [14], Hopfield [15]. We'll discuss the driven proofreading mechanisms previewed in this lecture note in the next lecture in which we'll discuss the inner environment of cells (including intracellular signaling).

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